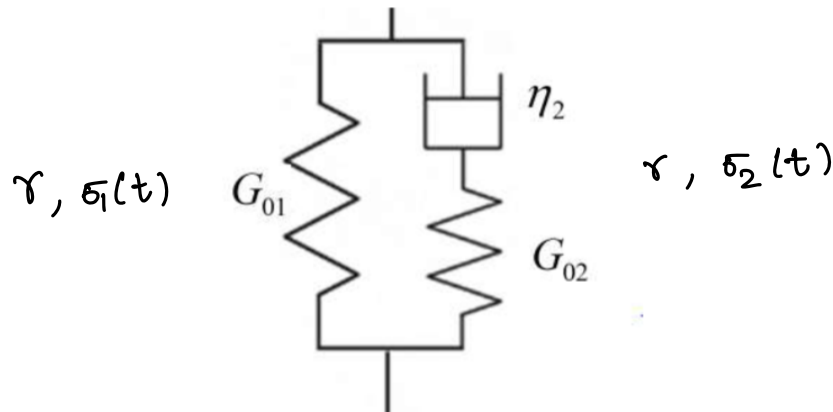


Assignment - 1

2. Consider a mechanistic rheological model shown in the figure below.



a) Obtain the constitutive equation for the abovementioned mechanistic model. The constitutive equation is defined as the relationship between stress (σ), strain (γ) and their time derivatives ($\dot{\sigma}, \dot{\gamma}$). The model parameters are: G_{01} , G_{02} , and η_2 .

b) The model undergoes a stress relaxation test, wherein a constant strain field, $\gamma = \gamma_0$ is imposed on the above system for $t \geq 0$. Solve the constitutive equation obtained in part (a) for step strain test and obtain stress (σ) as a function of time, plot $\sigma(t)$ as a function of t also.

Sol'n :

a)

$$\sigma(t) = \sigma_1 + \sigma_2$$

$$\gamma = \gamma_{e2} + \gamma_{v2}$$

$$\sigma_1 = G_{01} \gamma$$

$$\gamma = \gamma_{e2} + \gamma_{v2}$$

$$\dot{\gamma} = \dot{\gamma}_{e2} + \dot{\gamma}_{v2}$$

$$\dot{\gamma} = \frac{\dot{\sigma}_2}{G_{02}} + \frac{\sigma_2}{\eta_2}$$

$$\sigma_2 = \sigma - \sigma_1$$

$$\sigma_2 = \sigma - G_{01} \gamma$$

$$\dot{\sigma}_2 = \dot{\sigma} - G_{01} \dot{\gamma}$$

substituting

$$\dot{\gamma} = \frac{\dot{\sigma} - G_{01} \dot{\gamma}}{G_{02}} + \frac{\sigma - G_{01} \gamma}{\eta_2}$$

$$\dot{\gamma} = \frac{\dot{\sigma}}{G_{02}} - \frac{G_{01}}{G_{02}} \dot{\gamma} + \frac{\sigma}{\eta_2} - \frac{G_{01}}{\eta_2} \gamma$$

$$\frac{\sigma}{\eta_2} + \frac{\dot{\sigma}}{G_{02}} = \frac{G_{01}}{\eta_2} \gamma + \left(1 + \frac{G_{01}}{G_{02}}\right) \dot{\gamma} \rightarrow \text{CONSTITUTIVE Eq}^n$$

b) Now, step strain $\gamma = \gamma_0$, $\dot{\gamma} = 0$.

$$\Rightarrow \frac{\sigma}{\eta_2} + \frac{\dot{\sigma}}{G_{02}} = \frac{G_{01}}{\eta_2} \gamma_0$$

$$\Rightarrow \frac{d\sigma}{dt} + \frac{\sigma}{\eta_2/G_{02}} = \gamma_0 \left(\frac{G_{01} \cdot G_{02}}{\eta} \right)$$

$$\sigma \exp\left(\frac{t}{\tau_2}\right) \Big|_0^t = \frac{\gamma_0 \cdot G_{01} \cdot G_{02}}{\eta_2} \int_0^t \exp\left(\frac{t}{\tau_2}\right) dt$$

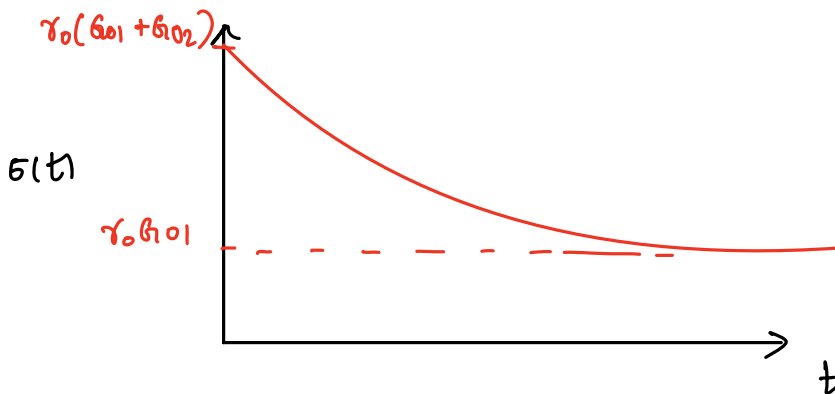
$$\sigma \exp\left(\frac{t}{\tau_2}\right) - \underbrace{\sigma(0)}_{\gamma_0(G_{01}+G_{02})} = \frac{\gamma_0 \cdot G_{01} \cdot \cancel{G_{02}}}{\cancel{\eta_2}} \times \frac{\eta_2}{\cancel{G_{02}}} \left[\exp\left(\frac{t}{\tau_2}\right) - 1 \right]$$

$$\sigma(t) \exp\left(\frac{t}{\tau_2}\right) - \gamma_0(G_{01}+G_{02}) = \gamma_0 G_{01} \left[\exp\left(\frac{t}{\tau_2}\right) - 1 \right]$$

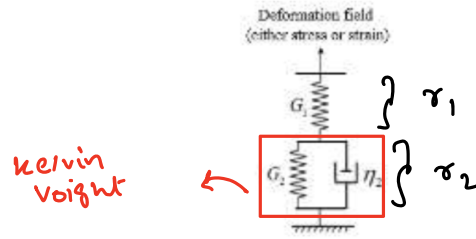
$$\sigma(t) \exp\left(\frac{t}{\tau_2}\right) = \gamma_0 G_{02} + \gamma_0 G_{01} \exp\left(\frac{t}{\tau_2}\right)$$

$$\sigma(t) = \gamma_0 G_{02} \exp\left(-\frac{t}{\tau_2}\right) + \gamma_0 G_{01}$$

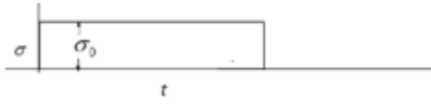
$$\sigma(t) = \gamma_0 G_{02} \exp\left(-\frac{t}{\tau_2}\right) + \gamma_0 G_{01}$$



3. Consider a mechanistic rheological model shown in the figure below.



The above model has been subjected to the constant stress (within linear regime) up to time t_1 , after which stress has been removed, and the strain starts getting recovered. Find out the expression of strain rate as a function of time during recovery step.

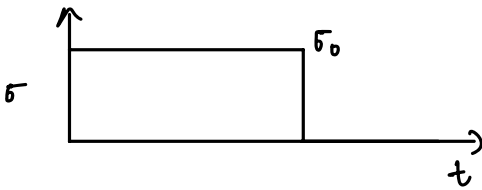


The constitutive equation is defined as the relationship between stress (σ), strain (γ) and their time derivatives ($\dot{\sigma}, \dot{\gamma}$).

The model parameters are: G_1 , G_2 , and η_2 .

- a) The model undergoes a step strain test, wherein a constant strain field, $\gamma = \gamma_0$ is imposed on the above system for $t \geq 0$. Solve the constitutive equation obtained in part (a) for step strain test and obtain stress (σ) as a function of time, plot $\sigma(t)$ as a function of t also.

Sol'n:



Let's arrive at the constitutive eqn first:

$$\sigma = \sigma_1 = \sigma_2$$

$$\gamma = \gamma_1 + \gamma_2$$

$$\sigma_1 = G_1 \gamma_1, \quad \sigma_2 = \sigma_2 e + \sigma_2 \phi = \gamma_2 G_2 + \dot{\gamma}_2 \eta_2$$

$$\Rightarrow \dot{\gamma} = \dot{\gamma}_1 + \dot{\gamma}_2$$

$$\dot{\gamma} = \frac{\dot{\sigma}}{G_1} + \frac{\sigma - \gamma_2 G_2}{\eta_2}$$

$$\dot{\gamma} = \frac{\dot{\sigma}}{G_1} + \frac{\sigma}{\eta_2} - \frac{G_2}{\eta_2} (\gamma - \gamma_1)$$

$$\dot{\gamma} = \frac{\dot{\sigma}}{G_1} + \frac{\sigma}{\eta_2} - \frac{G_2}{\eta_2} \left(\gamma - \frac{\sigma}{G_1} \right)$$

$$\dot{\gamma} + \frac{\gamma}{\tau_2} = \frac{\dot{\sigma}}{G_1} + \frac{\sigma}{\eta_2} + \frac{G_2}{\eta_2 G_1} \sigma$$

Now, for constant step stress applied

$$\sigma = \sigma_0, \quad \dot{\epsilon} = 0$$

$$\dot{\gamma} + \frac{\gamma}{\tau_2} = \frac{1}{\eta_2} \left(1 + \frac{G_2}{G_1} \right) \sigma_0$$

$$\gamma \exp\left(\frac{t}{\tau_2}\right) \Big|_0^{t_1} = \frac{1}{\eta_2} \left(1 + \frac{G_2}{G_1} \right) \sigma_0 \int_0^{t_1} \exp\left(\frac{t}{\tau_2}\right) dt$$

$$\sigma(t_1) \exp\left(\frac{t_1}{\tau_2}\right) - \sigma_0 = \frac{1}{\cancel{\eta_2}} \left(1 + \frac{G_2}{G_1} \right) \sigma_0 \times \frac{\cancel{\eta_2}}{G_2} \left[\exp\left(\frac{t_1}{\tau_2}\right) - 1 \right]$$

$$\sigma(t_1) \exp\left(\frac{t_1}{\tau_2}\right) - \sigma_0 = \sigma_0 \left(\frac{1}{G_2} + \frac{1}{G_1} \right) \left[\exp\left(\frac{t_1}{\tau_2}\right) - 1 \right]$$

initially all strain will be stored
in G_1 spring. $\Rightarrow \sigma_0 = G_1 \sigma$

$$\Rightarrow \gamma_0 = \sigma_0 / G_1$$

$$\sigma(t_1) \exp\left(\frac{t_1}{\tau_2}\right) - \frac{\sigma_0}{\cancel{G_1}} = \sigma_0 \left(\frac{1}{G_2} + \frac{1}{G_1} \right) \exp\left(\frac{t_1}{\tau_2}\right) - \frac{\sigma_0}{G_2} - \frac{\sigma_0}{\cancel{G_1}}$$

$$\sigma(t_1) \exp\left(\frac{t_1}{\tau_2}\right) = \frac{\sigma_0}{G_2} \left[\exp\left(\frac{t_1}{\tau_2}\right) - 1 \right] + \frac{\sigma_0}{G_1} \exp\left(\frac{t_1}{\tau_2}\right)$$

$$\sigma(t_1) = \frac{\sigma_0}{G_2} \left(1 - \exp\left(-\frac{t_1}{\tau_2}\right) \right) + \frac{\sigma_0}{G_1}$$

we can also arrive at this directly.

$$\sigma(t_1) = \underbrace{\frac{\sigma_0}{G_2} \left(1 - \exp\left(-\frac{t}{\tau_2}\right) \right)}_{\text{kelvin-Voigt}} + \underbrace{\frac{\sigma_0}{G_1}}_{\text{spring 1}}$$

Now after $(t > t_1)$, $\sigma = 0$, $\dot{\epsilon} = 0$

$$\dot{\gamma} + \frac{\gamma}{\tau} = 0$$

$$\frac{dr}{dt} = -\frac{r}{\tau_2} \Rightarrow \int_{t_1}^t \frac{dr}{r} = \int_{t_1}^t -\frac{dt}{\tau_2}$$

$$\ln \left(\frac{r_t}{r_{t_1}} \right) = \frac{t_1}{\tau_2} - \frac{t}{\tau_2}$$

$$r_t = r_{t_1} \exp \left(-\frac{(t - t_1)}{\tau_2} \right)$$

a)

$$\cancel{\dot{\sigma}} + \frac{\sigma}{\tau_2} = \frac{\dot{\sigma}}{G_1} + \frac{\sigma}{\eta_2} + \frac{G_2}{\eta_2 G_1} \sigma$$

$$\sigma = \sigma_0, \quad \dot{\sigma} = 0$$

$$\sigma \left(\frac{1}{\eta_2} + \frac{1}{\eta_2} \frac{G_2}{G_1} \right) + \frac{\dot{\sigma}}{G_1} = \frac{\sigma_0}{\tau_2}$$

$$\dot{\sigma} + \sigma \left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) = \frac{G_1 \sigma_0}{\tau_2}$$

$$\sigma \exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] \Bigg|_0^t = \frac{G_1 \sigma_0}{\tau_2} \int_0^t \exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] dt$$

$$\sigma(t) \exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] - \underbrace{\sigma(0)}_{\sigma_0 G_1} = \frac{G_1 \sigma_0}{\tau_2} \frac{\eta_2}{G_1 + G_2} \left[\exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] - 1 \right]$$

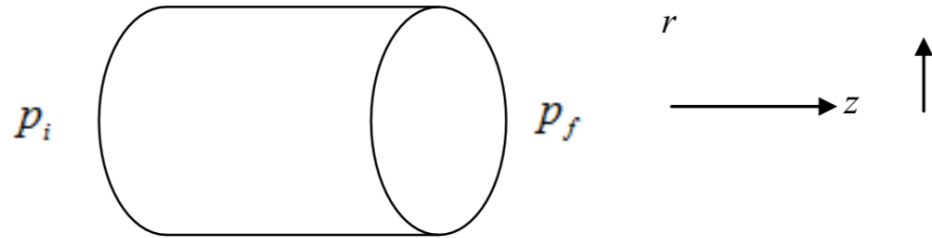
$$\sigma(t) \exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] - \sigma_0 G_1 = \sigma_0 \frac{G_1 G_2}{G_1 + G_2} \left[\exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] - \frac{\sigma_0 G_1 G_2}{G_1 + G_2} \right]$$

$$\sigma(t) \exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] = \sigma_0 \frac{G_1 G_2}{G_1 + G_2} \left[\exp \left[\left(\frac{G_1}{\eta_2} + \frac{G_2}{\eta_2} \right) t \right] - \frac{\cancel{\sigma_0 G_1 G_2} + \sigma_0 G_1^2 + \cancel{\sigma_0 G_2 \eta_2}}{G_1 + G_2} \right]$$

$$\sigma(t) = \frac{\sigma_0 G_1 G_2}{G_1 + G_2} - \frac{\sigma_0 G_1^2}{G_1 + G_2} \exp \left[- \left(\frac{G_1}{\eta_1} + \frac{G_2}{\eta_2} \right) t \right]$$

$$\sigma(t) = \frac{\sigma_0 G_1}{G_1 + G_2} \left[G_2 - G_1 \exp \left[- \left(\frac{G_1}{\eta_1} + \frac{G_2}{\eta_2} \right) t \right] \right]$$

1. Consider a pressure driven flow (linear pressure gradient, $\frac{-dp}{dz} = \frac{p_i - p_f}{L}$) through a cylindrical pipe of length L and radius r . Find out the steady state velocity profile ($V_z(r)$) for Newtonian as well as Power law fluid. (Assume fully developed flow).



Hint: Use Cauchy Momentum Balance in cylindrical coordinate system (given below). The

constitutive equation for Newtonian fluid: $\tau_{rz} = \eta \frac{\partial V_z}{\partial r}$, and for Power law fluid: $\tau_{rz} = \eta \left(\frac{\partial V_z}{\partial r} \right)^n$.

Sol'n :

Assignment - 2

1. Prove that G' and G'' as a function of ω are given in terms of relaxation modulus $G(t)$.

$$G'(\omega) = \omega \int_0^{\infty} G(a) (\sin \omega a) da$$

$$G''(\omega) = \omega \int_0^{\infty} G(a) (\cos \omega a) da$$

Sol'n: We can use BSP

$$\sigma(t) = \int_{-\infty}^t h(t-t') \dot{\gamma}(t') dt'$$

$$\begin{aligned} t-t' &= s \\ -dt' &= ds \end{aligned}$$

$$\sigma(t) = - \int_0^{\infty} h(s) \dot{\gamma}(t-s) ds$$

$$\gamma(t) = \gamma_0 \sin \omega t$$

$$\gamma(t-s) = \gamma_0 \sin(\omega t - \omega s)$$

$$\frac{d\gamma(t-s)}{ds} = -\omega \gamma_0 \cos(\omega t - \omega s)$$

$$\sigma(t) = \int_0^{\infty} h(s) \omega \gamma_0 \cos(\omega t - \omega s) ds$$

$$\sigma(t) = \omega \gamma_0 \int_0^{\infty} h(s) [\cos \omega t \cos \omega s + \sin \omega t \sin \omega s] ds$$

$$\frac{\sigma(t)}{\gamma_0} = \underbrace{\omega \cos \omega t \int_0^{\infty} h(s) \cos \omega s ds}_{\text{out of phase with } \sigma} + \underbrace{\omega \sin \omega t \int_0^{\infty} h(s) \sin \omega s ds}_{\text{in-phase with } \sigma}$$

$$\Downarrow$$

$$G''(\omega) = \omega \int_0^{\infty} G(s) \cos \omega s \, ds$$

$$\Downarrow$$

$$G'(\omega) = \omega \int_0^{\infty} G(s) \sin \omega s \, ds$$

2. Using the relaxation modulus for single mode Maxwell model ($G(t) = G_0 e^{-t/\tau_0}$), prove that G' and G'' are given by,

$$G' = \frac{G_0 \omega^2 \tau_0^2}{1 + \omega^2 \tau_0^2} \quad ; \quad G'' = \frac{G_0 \omega \tau_0}{1 + \omega^2 \tau_0^2}$$

Sol'n:

$$G^* = G' + i G''$$

$$G^* = \frac{\Delta \sigma}{\Delta \tau}$$

$$G(t) = G_0 e^{-t/\tau_0}$$

Using previous results

$$G' = \omega \int_0^{\infty} G(s) \sin \omega s \, ds$$

$$G' = \omega \int_0^{\infty} G_0 e^{-s/\tau_0} \sin \omega s \, ds$$

$$\frac{G'}{\omega G_0} = \int_0^{\infty} e^{-s/\tau_0} \sin \omega s \, ds = I$$

Applying by parts:-

$$\frac{G'}{\omega G_0} = I = \left[e^{-s/\tau_0} \frac{\cos \omega s}{\omega} \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{e^{-s/\tau_0}}{\tau_0} \right) \frac{\cos \omega s}{\omega} \, ds$$

$$\frac{G'}{\omega G_0} = I = \frac{1}{\omega} - \frac{1}{\omega \tau_0} \int_0^{\infty} e^{-s/\tau_0} \cos \omega s \, ds$$

Again applying by parts:-

$$\frac{G'}{\omega G_0} = I = \frac{1}{\omega} - \frac{1}{\omega \tau_0} \left[\left[\frac{e^{-s/\tau_0} \sin \omega s}{\omega} \right]_0^{\infty} - \int_0^{\infty} \left(-\frac{1}{\tau_0} e^{-s/\tau_0} \right) \left(\frac{\sin \omega s}{\omega} \right) \, ds \right]$$

$$\frac{G'}{\omega G_0} = I = \frac{1}{\omega} - \frac{1}{\omega \tau_0} \left[\frac{1}{\omega \tau_0} \int_0^{\omega \tau_0} e^{-s/\tau_0} \sin \omega s \, ds \right]$$

$$I = \frac{1}{\omega} - \frac{1}{\omega^2 \tau_0^2} I$$

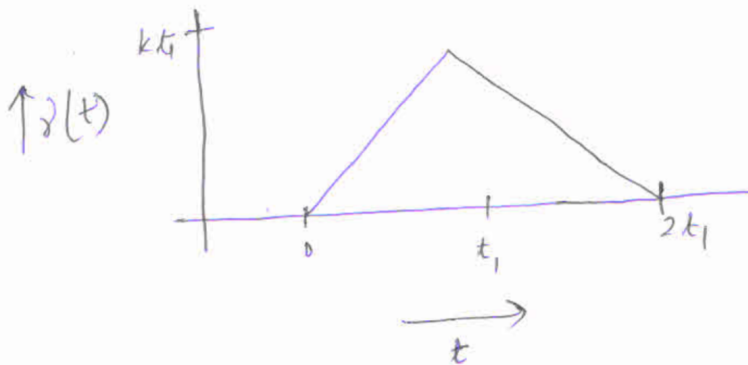
$$I \left(1 + \frac{1}{\omega^2 \tau_0^2} \right) = \frac{1}{\omega}$$

$$I = \frac{1}{\omega} \frac{\omega^2 \tau_0^2}{\omega^2 \tau_0^2 + 1} \quad \Rightarrow \quad G' = \frac{G_0 \omega^2 \tau_0^2}{\omega^2 \tau_0^2 + 1}$$

$$G' = \frac{G_0 \omega^2 \tau_0^2}{\omega^2 \tau_0^2 + 1}$$

similarly we can prove for G'' .

3. The relaxation modulus of a material is given by $G(t) = G_1 e^{-t/\tau_1} + G_2 e^{-t/\tau_2}$. Find out the stress for $t > 2t_1$ for the strain profile shown below.



Sol'n:

$$\sigma(t) = \int_{-\infty}^t G(t-t') \dot{\gamma}(t') dt'$$

$$\sigma(t) = \int_0^{t_1} G(t-t') \dot{\gamma}(t') dt' + \int_{t_1}^{2t_1} G(t-t') \dot{\gamma}(t') dt' + \int_{2t_1}^t G(t-t') \dot{\gamma}(t') dt'$$

$$\sigma(t) = \int_0^{t_1} \left[G_1 e^{-\frac{(t-t')}{\tau_1}} + G_2 e^{-\frac{(t-t')}{\tau_2}} \right] k dt' - \int_{t_1}^{2t_1} \left[G_1 e^{-\frac{(t-t')}{\tau_1}} + G_2 e^{-\frac{(t-t')}{\tau_2}} \right] k dt'$$

$$\sigma(t) \Rightarrow k \left[-\frac{G_1}{\tau_1} e^{-\frac{(t-t')}{\tau_1}} - \frac{G_2}{\tau_2} e^{-\frac{(t-t')}{\tau_2}} \right]_0^{t_1} - k \left[-\frac{G_1}{\tau_1} e^{-\frac{(t-t')}{\tau_1}} - \frac{G_2}{\tau_2} e^{-\frac{(t-t')}{\tau_2}} \right]_{t_1}^{2t_1}$$

$$\sigma(t) \Rightarrow k \left[-\frac{G_1}{\tau_1} e^{-t/\tau_1} - \frac{G_2}{\tau_2} e^{-t/\tau_2} + \cancel{\frac{G_1}{\tau_1} e^{-\frac{(t-t_1)}{\tau_1}}} + \cancel{\frac{G_2}{\tau_2} e^{-\frac{(t-t_1)}{\tau_2}}} \right]$$

$$- k \left[-\frac{G_1}{\tau_1} e^{-\frac{(t-2t_1)}{\tau_1}} - \frac{G_2}{\tau_2} e^{-\frac{(t-2t_1)}{\tau_2}} + \cancel{\frac{G_1}{\tau_1} e^{-\frac{(t-t_1)}{\tau_1}}} + \cancel{\frac{G_2}{\tau_2} e^{-\frac{(t-t_1)}{\tau_2}}} \right]$$

$$\Rightarrow \frac{kG_1}{\tau_1} \left[e^{-\frac{(t-2t_1)}{\tau_1}} - e^{-t/\tau_1} \right] + \frac{kG_2}{\tau_2} \left[e^{-\frac{(t-2t_1)}{\tau_2}} - e^{-t/\tau_2} \right]$$

$$\Rightarrow \frac{kG_1}{\tau_1} e^{-t/\tau_1} \left[e^{2t_1/\tau_1} - 1 \right] + \frac{kG_2}{\tau_2} e^{-t/\tau_2} \left[e^{2t_1/\tau_2} - 1 \right]$$

4. For multimode Maxwell Mode frame that overall viscosity (η) is given by $\eta = \sum_{i=1}^n G_{oi} \tau_{oi}$

Sol'n:

$$\eta = \int_0^{\infty} h(s) ds$$

$$\sigma(t) = \sum \sigma_i(t)$$

$$\sigma(t) = \sum \tau_0 G_{oi} e^{-t/\tau_i}$$

$$\frac{\sigma(t)}{\tau_0} = \sum G_{oi} e^{-t/\tau_i} \Rightarrow h(s) = \sum h_{oi}(s)$$

$$\therefore \eta = \int_0^{\infty} \sum_{i=1}^n h_{oi}(s) ds$$

$$\eta \Rightarrow \sum_{i=1}^n \int_0^{\infty} h_{oi}(s) ds \Rightarrow \sum_{i=1}^n \int_0^{\infty} G_{oi} e^{-s/\tau_{oi}} ds$$

$$\eta = \sum_{i=1}^n [-G_{0i} \tau_{0i} e^{-s/\tau_{0i}}]_0^{\infty}$$

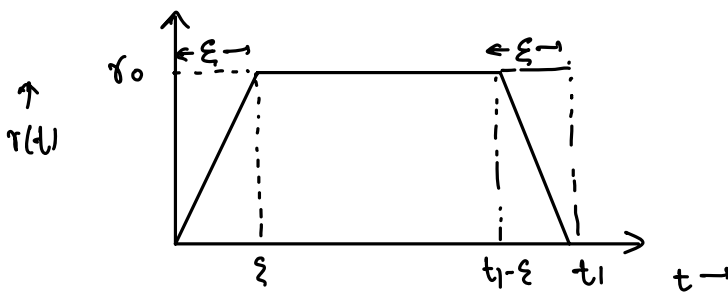
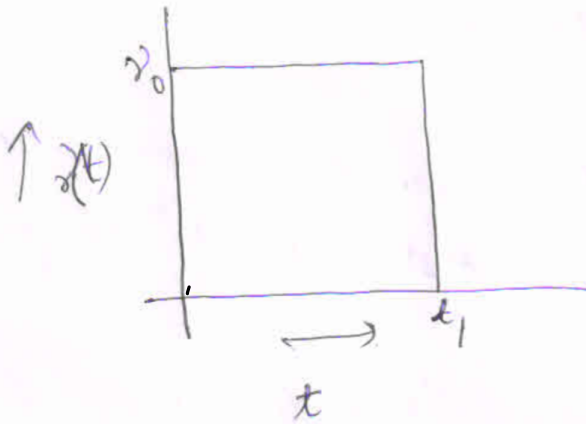
$$\boxed{\eta \Rightarrow \sum_{i=1}^n G_{0i} \tau_{0i}}$$

5. The relaxation modulus of a material is given by,

$$G(t) = G_0 e^{-t/\tau}$$

Find out the stress at time $t > t_1$ for the strain profile shown below -

$\Rightarrow$ If we don't consider ε we get $\sigma(t) = 0$



$$\sigma(t) = \int_{-\infty}^t G(t-t') \dot{\gamma}(t') dt'$$

$$\Rightarrow \int_0^{\varepsilon} G(t-t') \frac{\sigma_0}{\varepsilon} dt' - \int_{t_1-\varepsilon}^{t_1} G(t-t') \frac{\sigma_0}{\varepsilon} dt'$$

$$\sigma(t) \Rightarrow \lim_{\varepsilon \rightarrow 0} \int_0^{\varepsilon} G(t-t') \sigma_0 dt - \lim_{\varepsilon \rightarrow 0} \int_{t_1-\varepsilon}^{t_1} G(t-t') \sigma_0 dt'$$

$$\sigma(t) \Rightarrow \frac{G(t-\xi) \tau_0}{1} - \frac{G(t-t_1+\xi) \tau_0}{1}$$

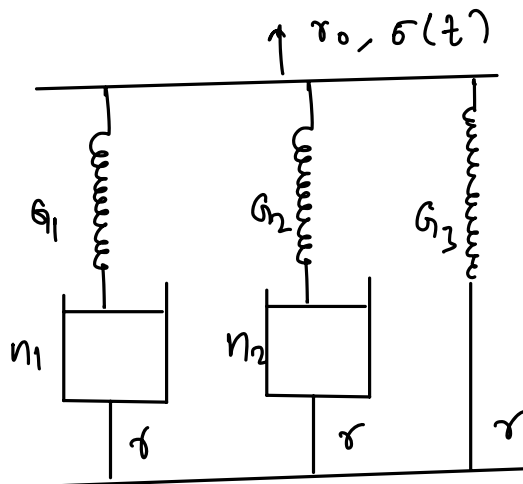
$$\sigma(t) \Rightarrow G(t) \tau_0 - G(t-t_1) \tau_0$$

$$\Rightarrow \tau_0 [G(t) - G(t-t_1)]$$

1. A polymer-nanocomposite when subjected to a step strain, the relaxation modulus shows two step exponential increase with an equilibrium modulus value, Answer the following:

- Suggest a mechanical model (consist of spring and dashpot elements), which can capture the above mentioned experimental behaviour.
- Derive the constitutive equation for the above proposed mechanical model. (assume your own model parameters)
- Solve your proposed mechanical model for step strain test and show that relaxation modulus indeed show shows two step exponential decay with an equilibrium modulus

$$a) G(t) = G_1 e^{-t/\tau_1} + G_2 e^{-t/\tau_2} + G_3 \infty$$



$$b) \sigma = \sigma_1 + \sigma_2 + \sigma_3 \quad \text{--- (1)}$$

$$\sigma_3 = G_3 \tau$$

$$\tau = \tau_2 e + \tau_2 v$$

$$\dot{\tau} = \dot{\tau}_2 e + \dot{\tau}_2 v$$

$$= \frac{\dot{\sigma}_2}{G_2} + \frac{\dot{\sigma}_2}{\eta_2}$$

$$\dot{\tau} = \frac{\dot{\sigma}_1}{G_1} + \frac{\dot{\sigma}_1}{\eta_1}$$

$$\dot{\sigma} = \dot{\sigma}_1 + \dot{\sigma}_2 + \dot{\sigma}_3 \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2}$$

$$\frac{\sigma}{\eta_1} + \frac{\dot{\sigma}}{G_1} = \frac{\sigma_1}{\eta_1} + \frac{\sigma_2}{\eta_1} + \frac{\sigma_3}{\eta_1} + \frac{\dot{\sigma}_1}{G_1} + \frac{\dot{\sigma}_2}{G_1} + \frac{\dot{\sigma}_3}{G_1}$$

$$\frac{\sigma}{\eta_1} + \frac{\dot{\sigma}}{G_1} = \dot{\gamma} + \frac{G_3 \sigma}{\eta_1} + \frac{G_3 \dot{\sigma}}{G_1} + \frac{\sigma_2}{\eta_1} + \frac{\dot{\sigma}_2}{G_1}$$

$$\frac{\sigma}{\eta_1} + \frac{\dot{\sigma}}{G_1} - \dot{\gamma} - \frac{G_3 \sigma}{\eta_1} - \frac{G_3 \dot{\sigma}}{G_1} = \frac{\sigma_2}{\eta_1} + \frac{\dot{\sigma}_2}{G_1}$$

$$p = \frac{\sigma_2}{\eta_1} + \frac{\dot{\sigma}_2}{G_1}$$

$$p = \frac{\sigma_2}{\eta_1} + \frac{G_2}{G_1} (\dot{\sigma} - \frac{\sigma_2}{\eta_2})$$

$$p = \frac{\sigma_2}{\eta_1} + \frac{G_2 \dot{\sigma}}{G_1} - \sigma_2 \frac{G_2}{G_1 \eta_2}$$

$$\frac{p - \frac{G_2 \dot{\sigma}}{G_1}}{\frac{1}{\eta_1} - \frac{G_2}{G_1 \eta_2}} = \sigma_2$$

calculate $\dot{\sigma}_2$ & plug $\dot{\sigma}_2$, σ_2 into main equation,

solve further $\gamma = \gamma_0$, $\dot{\gamma} = \ddot{\gamma} = 0$.